Model under test: Track 1 baseline CNN (CIFAR-10 animals: bird, cat, deer, dog, frog, horse; 67.08% test accuracy).

1. Experiment plan

Question: does the baseline recognise the animal, or does it lean on the background and colour? The Track 2 failure analysis hints at the second: uncertain images were pushed into "frog" and "horse" (e.g. 247 cats and 157 birds predicted as frog), often on mottled green/brown backgrounds.

Task	**Output**
Fix the thresholds in section 2 and a random seed. Build object masks (centre-region proxy) and hand-check 50.	Masks and a quality note
Generate the modified test sets (section 3) and run the baseline on each.	Accuracy per class, per test set
Compute the metrics in section 2. Run Grad-CAM and measure how much attention falls on the background.	Metrics table, 10 example heatmaps
Train a second model with a deliberately planted shortcut and check that our metrics detect it (positive control).	Control result
Ablate the model parts tied to the shortcut and see whether predictions recover. Write up findings.	1-page summary

2. What "shortcut reliance" means, operationally

A shortcut is any image feature that predicts the label in the training data but is not the animal itself: background, colour palette, texture, or a planted marker. The model relies on a shortcut when changing only that feature changes the prediction while the animal stays the same.

Conflict drop	Accuracy on normal images minus accuracy on the same animals with the background swapped to another class's	Drop is more than 10 points
Background-only accuracy	Accuracy when the animal is masked out and only the background remains	Well above chance (16.7%); we flag above 30%
Object-only retention	Accuracy with the background replaced by flat grey, divided by original accuracy	Below 0.8
Grad-CAM background share	Share of Grad-CAM attention that falls outside the animal	Larger than the background's share of the image
Intervention check	Prediction change after switching off the model parts most tied to the shortcut	Prediction moves back toward the true class

3. Images and classes to test
Images come from the CIFAR-10 test set. Per class, sample 100 correctly classified and 100 misclassified images (fixed seed).

Test set	**How it is built**	**What it tests**
Original	Unmodified images	Reference accuracy
Background-only	Animal masked out (grey fill)	Can the background alone predict the class?
Object-only	Background replaced by flat grey	Does the animal alone suffice?
Background swap	Animal pasted onto another class's background (e.g. horse on a green pond scene)	Conflict drop
Greyscale	Colour removed, shape and texture kept	Is colour the shortcut?
Colour shuffle	RGB channels permuted	Colour vs shape
Planted shortcut	Second model trained with a cue tied to one class (e.g. coloured corner patch on 90% of frogs); patch added to other classes at test time	Positive control: do our metrics detect a known shortcut?

Classes:

1. Frog and horse: (recall 0.93, precision 0.57 and 0.61).
2. Cat and dog: lowest recall (0.42, 0.45), and most of their errors are not cat to dog, so background is a likely cause.
3. Bird (recall 0.59): wide background variety (sky, branches, water).
4. Deer: control class; it nearly always appears on grass, so it is the most likely to show a legitimate-looking background cue.

Error pairs to examine: cat to frog, dog to horse, bird to frog, deer to frog, with cat to dog as a control for genuine visual similarity.